1. Purpose

The purpose of this Validation Plan is to define the strategy and activities required to validate the Cloud-Hosted Laboratory Sample Tracking System. To ensure it is fit for intended use and compliant with applicable regulatory requirements including 21 CFR Part 11, GAMP 5, and ALCOA+ data integrity principles.

2. Scope of Validation

This validation applies to the cloud-hosted Laboratory Sample Tracking System, including user access control, audit trail functionality, sample data entry, reporting, and data export features. Infrastructure components outside the application boundaries, such as laboratory instruments and enterprise network systems, are excluded from this validation scope.

2.1 In-Scope

1. Sample Tracking Application
2. User authentication and role-based access
3. Audit trail functionality
4. Sample data entry and reporting
5. Data export functionality

Hosted in Microsoft Azure environment.

2.2 Out of Scope

1. Instrument calibration
2. Network infrastructure outside application scope

3. Definitions

URS | User Requirements Specification
IQ | Installation Qualification
OQ | Operational Qualification
PQ | Performance Qualification
ALCOA+ | is a framework for ensuring data integrity, emphasizing that data must be Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, and Available
21 CFR Part 11 | is an FDA regulation that sets the standard for electronic records and electronic signatures ensuring they are trustworthy, reliable, and equivalent to paper records and handwritten signatures
GAMP 5 | a set of guidelines developed by International Society for Pharmaceutical Engineering (ISPE). It provides a framework to ensure the quality, safety, and compliance of automated systems

in the pharmaceutical and like sciences industries. GAMP 5 emphasizes a risk-based approach to computer system validation, helping professionals manage automated systems throughout their lifecycle

4. System Description

The Sample Tracking System is a configurable web-based application hosted by Microsoft Azure used to:

1. Track laboratory samples
2. Assign analysts
3. Record test results
4. Maintain audit trail of changes
5. Export reports to Excel

4. Regulatory Requirements

Validation will align with:

1. 21 CFR Part 11
2. FDA Data Integrity Guidance
3. GAMP 5 Risk-Based Validation
4. ALCOA+ Data Integrity Principles
5. Company SOPs and Quality Manual

5. Validation Strategy

A risk-based validation approach will be used per GAMP 5.
System classified as GAMP Category 4 – Configurable Software.

5.1 Validation. Lifecycles will include:

1 Validation Plan
2. URS
3. Risk Assessment
4. Traceability Matrix
5. IQ
6. OQ
7. PQ
8. Validation Summary Report

6. Roles & Responsibilities

Validation Lead – Create validation documents
QA Reviewer – Review & approve validation
IT Support – Configure system environment
System Owner – Define user requirements
Testers – Execute IQ/OQ/PQ protocols

7. Deliverables

1. Validation Plan
2. URS
3. Risk Assessment
4. Traceability Matrix
5. IQ Protocol
6. OQ Protocol
7. PQ Protocol
8. Validation Summary Report

8. Acceptance Criteria

System will be considered validated when:
1. All test cases executed
2. No critical deviations remain open
3. High-risk functions verified
4. QA approval obtained

9. Deviation Handling

Any test failures will be documented as deviations. Root cause analysis will be performed using CAPA methodology and retesting conducted as needed.

10. Change Control

Any system changes post-validation will follow formal change control procedures with impact assessment to determine need for re-validation.

11. Document Approval

Validation Lead – Brian Alston
QA Reviewer – TBD
System Owner – TBD